

Line-by-Line Mathematical Guide for HAETAE_Reductions.ec

Generated HAETAE EasyCrypt proof guide

2026-05-11

File Role

Concrete loss terms and arithmetic normalization.

How To Read This Script

Read this file as the real-arithmetic ledger. It collects the bound expressions and proves they can be grouped in the form consumed by the top-level security theorem.

Line-by-Line Mathematical Reading

Each row preserves one EasyCrypt source line and gives the intended mathematical or proof-script reading. Blank rows are kept because they mark proof structure in long EasyCrypt developments.

Line	EasyCrypt source	Mathematical reading
1	<code>require import AllCore Real RealExp StdOrder.</code>	Imports AllCore Real RealExp StdOrder. Mathematically, this exposes earlier carrier sets, operations, games, and lemmas that the current file may cite.
2	<code>require import HAETAE_Events.</code>	Imports HAETAE_Events. Mathematically, this exposes earlier carrier sets, operations, games, and lemmas that the current file may cite.
3	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
4	<code>theory HAETAE_Reductions.</code>	Begins the namespace HAETAE_Reductions, grouping the declarations as one checked theory.
5	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
6	<code>import HAETAE_Events.</code>	Opens HAETAE_Events so its names can be used without qualification in the following proof.
7	<code>import RealOrder.</code>	Opens RealOrder so its names can be used without qualification in the following proof.
8	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
9	<code>op mlwe_hardness_term : real.</code>	Defines real-valued term mlwe_hardness_term, usually a loss, probability bound, or advantage expression.
10	<code>op module_sis_hardness_term : real.</code>	Defines real-valued term module_sis_hardness_term, usually a loss, probability bound, or advantage expression.
11	<code>op bimodal_to_msis_reduction_loss_term : real = 0%r.</code>	Defines real-valued term bimodal_to_msis_reduction_loss_term, usually a loss, probability bound, or advantage expression.
12	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
13	<code>op bimodal_selftarget_msis_hardness_term =</code>	Defines operation bimodal_selftarget_msis_hardness_term, a total mathematical function or abbreviation used by later declarations.
14	<code>module_sis_hardness_term + bimodal_to_msis_reduction_loss_term.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
15	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
16	op signature_query_count : real = 2%r.	Defines real-valued term signature_query_count, usually a loss, probability bound, or advantage expression.
17	op hash_query_count : real = 2%r.	Defines real-valued term hash_query_count, usually a loss, probability bound, or advantage expression.
18	op signature_query_budget_count : int = 2.	Defines integer-valued parameter signature_query_budget_count, used for dimensions, budgets, support sizes, or bounds.
19	op hash_query_budget_count : int = 2.	Defines integer-valued parameter hash_query_budget_count, used for dimensions, budgets, support sizes, or bounds.
20	op abort_probability : real = 0%r.	Defines real-valued term abort_probability, usually a loss, probability bound, or advantage expression.
21	op min_entropy_loss_factor : real = 2%r.	Defines real-valued term min_entropy_loss_factor, usually a loss, probability bound, or advantage expression.
22	op challenge_support_bits_lower_bound : int = 58.	Defines integer-valued parameter challenge_support_bits_lower_bound, used for dimensions, budgets, support sizes, or bounds.
23	op challenge_support_cardinality_lower_bound : real =	Defines real-valued term challenge_support_cardinality_lower_bound, usually a loss, probability bound, or advantage expression.
24	288230376151711744%r.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
25	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
26	op rom_hash_query_budget : real = hash_query_count.	Defines real-valued term rom_hash_query_budget, usually a loss, probability bound, or advantage expression.
27	op rom_signature_query_budget : real = signature_query_count.	Defines real-valued term rom_signature_query_budget, usually a loss, probability bound, or advantage expression.
28	op rom_total_query_budget =	Defines operation rom_total_query_budget, a total mathematical function or abbreviation used by later declarations.
29	rom_hash_query_budget + rom_signature_query_budget + 1%r.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
30	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
31	op counted_rom_collision_term =	Defines operation counted_rom_collision_term, a total mathematical function or abbreviation used by later declarations.
32	(rom_total_query_budget * rom_total_query_budget) /	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
33	challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
34	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
35	op counted_rom_prequery_term =	Defines operation counted_rom_prequery_term, a total mathematical function or abbreviation used by later declarations.
36	(rom_signature_query_budget * (rom_hash_query_budget + 1%r)) /	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
37	challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
38	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
39	op counted_rom_min_entropy_term =	Defines operation counted_rom_min_entropy_term, a total mathematical function or abbreviation used by later declarations.
40	(rom_signature_query_budget * (rom_hash_query_budget + 1%r)) /	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
41	challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
42	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
43	op counted_rom_prequery_reprogramming_term =	Defines operation counted_rom_prequery_reprogramming_term, a total mathematical function or abbreviation used by later declarations.
44	counted_rom_collision_term + counted_rom_prequery_term.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
45	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
46	op counted_rom_programming_loss_term =	Defines operation counted_rom_programming_loss_term, a total mathematical function or abbreviation used by later declarations.
47	counted_rom_collision_term +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
48	counted_rom_prequery_term +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
49	counted_rom_min_entropy_term.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
50	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
51	op signing_query_scale =	Defines operation signing_query_scale, a total mathematical function or abbreviation used by later declarations.
52	signature_query_count / (1%r - abort_probability).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
53	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
54	op fs_with_aborts_reprogramming_term =	Defines operation fs_with_aborts_reprogramming_term, a total mathematical function or abbreviation used by later declarations.
55	sqrt signing_query_scale *	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
56	(hash_query_count + 1%r + sqrt signing_query_scale).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
57	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
58	op fs_with_aborts_min_entropy_term =	Defines operation fs_with_aborts_min_entropy_term, a total mathematical function or abbreviation used by later declarations.
59	min_entropy_loss_factor *	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
60	(signing_query_scale * (hash_query_count + 1%r)).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
61	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
62	op rejection_sampling_loss_term =	Defines operation rejection_sampling_loss_term, a total mathematical function or abbreviation used by later declarations.
63	fs_with_aborts_reprogramming_term + fs_with_aborts_min_entropy_term.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
64	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
65	op haetae_euf_bound =	Defines operation haetae_euf_bound, a total mathematical function or abbreviation used by later declarations.
66	mlwe_hardness_term +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
67	bimodal_selftarget_msis_hardness_term +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
68	rejection_sampling_loss_term.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
69	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
70	op haetae_euf_counted_rom_bound =	Defines operation haetae_euf_counted_rom_bound, a total mathematical function or abbreviation used by later declarations.
71	mlwe_hardness_term +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
72	bimodal_selftarget_msis_hardness_term +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
73	counted_rom_programming_loss_term.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
74	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
75	lemma haetae_euf_boundE :	States checked lemma haetae_euf_boundE. The following proof script must derive this proposition from prior definitions and lemmas.
76	haetae_euf_bound =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
77	mlwe_hardness_term +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
78	module_sis_hardness_term +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
79	bimodal_to_msis_reduction_loss_term +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
80	fs_with_aborts_reprogramming_term +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
81	<code>fs_with_aborts_min_entropy_term.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
82	<code>proof.</code>	Begins the proof script for the preceding proposition.
83	<code>by rewrite /haetae_euf_bound /bimodal_selftarget_msis_hardness_term</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
84	<code>/rejection_sampling_loss_term; ring.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
85	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
86	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
87	<code>lemma haetae_euf_bound_groupedE :</code>	States checked lemma <code>haetae_euf_bound_groupedE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
88	<code>haetae_euf_bound =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
89	<code>(mlwe_hardness_term +</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
90	<code>(module_sis_hardness_term + bimodal_to_msis_reduction_loss_term)) +</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
91	<code>(fs_with_aborts_reprogramming_term +</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
92	<code>fs_with_aborts_min_entropy_term).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
93	<code>proof.</code>	Begins the proof script for the preceding proposition.
94	<code>by rewrite /haetae_euf_bound /bimodal_selftarget_msis_hardness_term</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
95	<code>/rejection_sampling_loss_term; ring.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
96	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
97	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
98	<code>lemma haetae_euf_bound_rejection_groupedE :</code>	States checked lemma <code>haetae_euf_bound_rejection_groupedE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
99	<code>haetae_euf_bound =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
100	<code>(mlwe_hardness_term +</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
101	<code>(module_sis_hardness_term + bimodal_to_msis_reduction_loss_term)) +</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
102	<code>rejection_sampling_loss_term.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
103	<code>proof.</code>	Begins the proof script for the preceding proposition.
104	<code>by rewrite /haetae_euf_bound /bimodal_selftarget_msis_hardness_term; ring.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
105	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
106	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
107	<code>lemma haetae_euf_counted_rom_bound_groupedE :</code>	States checked lemma <code>haetae_euf_counted_rom_bound_groupedE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
108	<code>haetae_euf_counted_rom_bound =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
109	<code>(mlwe_hardness_term +</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
110	<code>(module_sis_hardness_term + bimodal_to_msis_reduction_loss_term)) +</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
111	<code>counted_rom_programming_loss_term.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
112	<code>proof.</code>	Begins the proof script for the preceding proposition.
113	<code>by rewrite /haetae_euf_counted_rom_bound</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
114	<code>/bimodal_selftarget_msis_hardness_term; ring.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
115	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
116	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.

Line	EasyCrypt source	Mathematical reading
117	lemma signing_query_scaleE :	States checked lemma signing_query_scaleE. The following proof script must derive this proposition from prior definitions and lemmas.
118	signing_query_scale = signature_query_count.	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
119	proof.	Begins the proof script for the preceding proposition.
120	by rewrite /signing_query_scale /signature_query_count /abort_probability; ring.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
121	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
122	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
123	lemma signing_query_scale_nonnegative :	States checked lemma signing_query_scale_nonnegative. The following proof script must derive this proposition from prior definitions and lemmas.
124	0%r <= signing_query_scale.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
125	proof.	Begins the proof script for the preceding proposition.
126	by rewrite signing_query_scaleE /signature_query_count.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
127	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
128	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
129	lemma fs_with_aborts_reprogramming_term_nonnegative :	States checked lemma fs_with_aborts_reprogramming_term_nonnegative. The following proof script must derive this proposition from prior definitions and lemmas.
130	0%r <= fs_with_aborts_reprogramming_term.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
131	proof.	Begins the proof script for the preceding proposition.
132	rewrite /fs_with_aborts_reprogramming_term.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
133	apply mulr_ge0.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
134	+ by apply ge0_sqrt.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
135	rewrite addr_ge0.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
136	+ by rewrite addr_ge0 /hash_query_count.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
137	+ by apply ge0_sqrt.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
138	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
139	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
140	lemma fs_with_aborts_min_entropy_term_nonnegative :	States checked lemma fs_with_aborts_min_entropy_term_nonnegative. The following proof script must derive this proposition from prior definitions and lemmas.
141	0%r <= fs_with_aborts_min_entropy_term.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
142	proof.	Begins the proof script for the preceding proposition.
143	rewrite /fs_with_aborts_min_entropy_term.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
144	apply mulr_ge0.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
145	+ by rewrite /min_entropy_loss_factor.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
146	apply mulr_ge0.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
147	+ by apply signing_query_scale_nonnegative.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
148	by rewrite addr_ge0 /hash_query_count.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
149	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
150	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
151	lemma rom_hash_query_budget_nonnegative :	States checked lemma rom_hash_query_budget_nonnegative. The following proof script must derive this proposition from prior definitions and lemmas.
152	0%r <= rom_hash_query_budget.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.

Line	EasyCrypt source	Mathematical reading
153	proof. by rewrite /rom_hash_query_budget /hash_query_count. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
154	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
155	lemma rom_signature_query_budget_nonnegative :	States checked lemma rom_signature_query_budget_nonnegative. The following proof script must derive this proposition from prior definitions and lemmas.
156	0%r <= rom_signature_query_budget.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
157	proof. by rewrite /rom_signature_query_budget /signature_query_count. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
158	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
159	lemma hash_query_budget_countE :	States checked lemma hash_query_budget_countE. The following proof script must derive this proposition from prior definitions and lemmas.
160	hash_query_count = (hash_query_budget_count)%r.	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
161	proof. by rewrite /hash_query_count /hash_query_budget_count. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
162	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
163	lemma signature_query_budget_countE :	States checked lemma signature_query_budget_countE. The following proof script must derive this proposition from prior definitions and lemmas.
164	signature_query_count = (signature_query_budget_count)%r.	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
165	proof. by rewrite /signature_query_count /signature_query_budget_count. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
166	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
167	lemma hash_query_budget_count_nonnegative :	States checked lemma hash_query_budget_count_nonnegative. The following proof script must derive this proposition from prior definitions and lemmas.
168	0 <= hash_query_budget_count.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
169	proof. by rewrite /hash_query_budget_count. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
170	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
171	lemma signature_query_budget_count_nonnegative :	States checked lemma signature_query_budget_count_nonnegative. The following proof script must derive this proposition from prior definitions and lemmas.
172	0 <= signature_query_budget_count.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
173	proof. by rewrite /signature_query_budget_count. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
174	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
175	lemma min_entropy_loss_factor_nonnegative :	States checked lemma min_entropy_loss_factor_nonnegative. The following proof script must derive this proposition from prior definitions and lemmas.
176	0%r <= min_entropy_loss_factor.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
177	proof. by rewrite /min_entropy_loss_factor. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
178	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
179	lemma challenge_support_cardinality_lower_bound_positive :	States checked lemma challenge_support_cardinality_lower_bound_positive. The following proof script must derive this proposition from prior definitions and lemmas.
180	0%r < challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
181	proof. by rewrite /challenge_support_cardinality_lower_bound. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
182	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
183	lemma challenge_support_cardinality_lower_bound_nonnegative :	States checked lemma challenge_support_cardinality_lower_bound_nonnegative. The following proof script must derive this proposition from prior definitions and lemmas.

Line	EasyCrypt source	Mathematical reading
184	<code>0%r <= challenge_support_cardinality_lower_bound.</code>	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
185	<code>proof.</code>	Begins the proof script for the preceding proposition.
186	<code>by apply ltrW; apply challenge_support_cardinality_lower_bound_positive.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
187	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
188	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
189	<code>lemma rom_total_query_budget_nonnegative :</code>	States checked lemma <code>rom_total_query_budget_nonnegative</code> . The following proof script must derive this proposition from prior definitions and lemmas.
190	<code>0%r <= rom_total_query_budget.</code>	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
191	<code>proof.</code>	Begins the proof script for the preceding proposition.
192	<code>rewrite /rom_total_query_budget.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
193	<code>apply addr_ge0.</code>	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
194	<code>+ by apply addr_ge0;</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
195	<code>[apply rom_hash_query_budget_nonnegative</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
196	<code> apply rom_signature_query_budget_nonnegative].</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
197	<code>+ by [].</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
198	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
199	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
200	<code>lemma counted_rom_collision_term_nonnegative :</code>	States checked lemma <code>counted_rom_collision_term_nonnegative</code> . The following proof script must derive this proposition from prior definitions and lemmas.
201	<code>0%r <= counted_rom_collision_term.</code>	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
202	<code>proof.</code>	Begins the proof script for the preceding proposition.
203	<code>rewrite /counted_rom_collision_term.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
204	<code>apply divr_ge0.</code>	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
205	<code>+ by apply mulr_ge0; apply rom_total_query_budget_nonnegative.</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
206	<code>by apply challenge_support_cardinality_lower_bound_nonnegative.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
207	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
208	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
209	<code>lemma counted_rom_prequery_term_nonnegative :</code>	States checked lemma <code>counted_rom_prequery_term_nonnegative</code> . The following proof script must derive this proposition from prior definitions and lemmas.
210	<code>0%r <= counted_rom_prequery_term.</code>	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
211	<code>proof.</code>	Begins the proof script for the preceding proposition.
212	<code>rewrite /counted_rom_prequery_term.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
213	<code>apply divr_ge0.</code>	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
214	<code>+ apply mulr_ge0.</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
215	<code>+ by apply rom_signature_query_budget_nonnegative.</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
216	<code>by apply addr_ge0; [apply rom_hash_query_budget_nonnegative].</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
217	<code>by apply challenge_support_cardinality_lower_bound_nonnegative.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
218	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
219	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.

Line	EasyCrypt source	Mathematical reading
220	lemma counted_rom_min_entropy_term_nonnegative :	States checked lemma counted_rom_min_entropy_term_nonnegative. The following proof script must derive this proposition from prior definitions and lemmas.
221	0%r <= counted_rom_min_entropy_term.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
222	proof.	Begins the proof script for the preceding proposition.
223	rewrite /counted_rom_min_entropy_term.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
224	apply divr_ge0.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
225	+ apply mulr_ge0.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
226	+ by apply rom_signature_query_budget_nonnegative.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
227	by apply addr_ge0; [apply rom_hash_query_budget_nonnegative].	Discharges the current goal in one step using the cited simplification, lemma, or automation.
228	by apply challenge_support_cardinality_lower_bound_nonnegative.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
229	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
230	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
231	lemma counted_rom_prequery_reprogramming_term_nonnegative :	States checked lemma counted_rom_prequery_reprogramming_term_nonnegative. The following proof script must derive this proposition from prior definitions and lemmas.
232	0%r <= counted_rom_prequery_reprogramming_term.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
233	proof.	Begins the proof script for the preceding proposition.
234	rewrite /counted_rom_prequery_reprogramming_term.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
235	by apply addr_ge0;	Discharges the current goal in one step using the cited simplification, lemma, or automation.
236	[apply counted_rom_collision_term_nonnegative	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
237	apply counted_rom_prequery_term_nonnegative].	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
238	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
239	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
240	lemma counted_rom_programming_loss_term_nonnegative :	States checked lemma counted_rom_programming_loss_term_nonnegative. The following proof script must derive this proposition from prior definitions and lemmas.
241	0%r <= counted_rom_programming_loss_term.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
242	proof.	Begins the proof script for the preceding proposition.
243	rewrite /counted_rom_programming_loss_term.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
244	apply addr_ge0.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
245	+ by apply addr_ge0;	Solves a proof branch produced by a previous split, transitivity, or case analysis.
246	[apply counted_rom_collision_term_nonnegative	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
247	apply counted_rom_prequery_term_nonnegative].	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
248	+ by apply counted_rom_min_entropy_term_nonnegative.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
249	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
250	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
251	lemma counted_rom_collision_termE :	States checked lemma counted_rom_collision_termE. The following proof script must derive this proposition from prior definitions and lemmas.
252	counted_rom_collision_term =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
253	25%r / challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
254	proof.	Begins the proof script for the preceding proposition.

Line	EasyCrypt source	Mathematical reading
255	by rewrite /counted_rom_collision_term	Discharges the current goal in one step using the cited simplification, lemma, or automation.
256	/rom_total_query_budget	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
257	/rom_hash_query_budget	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
258	/rom_signature_query_budget	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
259	/hash_query_count	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
260	/signature_query_count; ring.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
261	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
262	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
263	lemma counted_rom_prequery_termE :	States checked lemma counted_rom_prequery_termE. The following proof script must derive this proposition from prior definitions and lemmas.
264	counted_rom_prequery_term =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
265	6%r / challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
266	proof.	Begins the proof script for the preceding proposition.
267	by rewrite /counted_rom_prequery_term	Discharges the current goal in one step using the cited simplification, lemma, or automation.
268	/rom_signature_query_budget	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
269	/rom_hash_query_budget	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
270	/hash_query_count	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
271	/signature_query_count; ring.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
272	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
273	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
274	lemma counted_rom_min_entropy_termE :	States checked lemma counted_rom_min_entropy_termE. The following proof script must derive this proposition from prior definitions and lemmas.
275	counted_rom_min_entropy_term =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
276	6%r / challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
277	proof.	Begins the proof script for the preceding proposition.
278	by rewrite /counted_rom_min_entropy_term	Discharges the current goal in one step using the cited simplification, lemma, or automation.
279	/rom_signature_query_budget	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
280	/rom_hash_query_budget	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
281	/hash_query_count	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
282	/signature_query_count; ring.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
283	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
284	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
285	lemma counted_rom_prequery_reprogramming_termE :	States checked lemma counted_rom_prequery_reprogramming_termE. The following proof script must derive this proposition from prior definitions and lemmas.
286	counted_rom_prequery_reprogramming_term =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
287	31%r / challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
288	proof.	Begins the proof script for the preceding proposition.
289	by rewrite /counted_rom_prequery_reprogramming_term	Discharges the current goal in one step using the cited simplification, lemma, or automation.
290	counted_rom_collision_termE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
291	counted_rom_prequery_termE; ring.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
292	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
293	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
294	lemma counted_rom_programming_loss_termE :	States checked lemma counted_rom_programming_loss_termE. The following proof script must derive this proposition from prior definitions and lemmas.
295	counted_rom_programming_loss_term =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
296	37%r / challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
297	proof.	Begins the proof script for the preceding proposition.
298	by rewrite /counted_rom_programming_loss_term	Discharges the current goal in one step using the cited simplification, lemma, or automation.
299	counted_rom_collision_termE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
300	counted_rom_prequery_termE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
301	counted_rom_min_entropy_termE; ring.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
302	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
303	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
304	lemma counted_rom_programming_loss_term_subunit :	States checked lemma counted_rom_programming_loss_term_subunit. The following proof script must derive this proposition from prior definitions and lemmas.
305	counted_rom_programming_loss_term < 1%r.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
306	proof.	Begins the proof script for the preceding proposition.
307	rewrite counted_rom_programming_loss_termE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
308	rewrite ltr_pdivr_mulr.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
309	+ by apply challenge_support_cardinality_lower_bound_positive.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
310	by rewrite /challenge_support_cardinality_lower_bound.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
311	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
312	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
313	lemma counted_rom_programming_loss_term_positive :	States checked lemma counted_rom_programming_loss_term_positive. The following proof script must derive this proposition from prior definitions and lemmas.
314	0%r < counted_rom_programming_loss_term.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
315	proof.	Begins the proof script for the preceding proposition.
316	rewrite counted_rom_programming_loss_termE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
317	apply divr_gt0.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
318	+ by [].	Solves a proof branch produced by a previous split, transitivity, or case analysis.
319	by apply challenge_support_cardinality_lower_bound_positive.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
320	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
321	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
322	lemma counted_rom_programming_loss_term_le1 :	States checked lemma counted_rom_programming_loss_term_le1. The following proof script must derive this proposition from prior definitions and lemmas.
323	1%r >= counted_rom_programming_loss_term.	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
324	proof.	Begins the proof script for the preceding proposition.
325	by apply ltrW; apply counted_rom_programming_loss_term_subunit.	Discharges the current goal in one step using the cited simplification, lemma, or automation.

Line	EasyCrypt source	Mathematical reading
326	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
327	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
328	lemma counted_rom_prequery_reprogramming_budget_numeratorE :	States checked lemma counted_rom_prequery_reprogramming_budget_numeratorE. The following proof script must derive this proposition from prior definitions and lemmas.
329	(rom_total_query_budget * rom_total_query_budget) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
330	(rom_signature_query_budget * (rom_hash_query_budget + 1%r)) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
331	31%r.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
332	proof.	Begins the proof script for the preceding proposition.
333	by rewrite /rom_total_query_budget	Discharges the current goal in one step using the cited simplification, lemma, or automation.
334	/rom_hash_query_budget	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
335	/rom_signature_query_budget	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
336	/hash_query_count	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
337	/signature_query_count; ring.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
338	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
339	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
340	lemma counted_rom_budget_numeratorE :	States checked lemma counted_rom_budget_numeratorE. The following proof script must derive this proposition from prior definitions and lemmas.
341	(rom_total_query_budget * rom_total_query_budget) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
342	(rom_signature_query_budget * (rom_hash_query_budget + 1%r)) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
343	(rom_signature_query_budget * (rom_hash_query_budget + 1%r)) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
344	37%r.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
345	proof.	Begins the proof script for the preceding proposition.
346	by rewrite /rom_total_query_budget	Discharges the current goal in one step using the cited simplification, lemma, or automation.
347	/rom_hash_query_budget	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
348	/rom_signature_query_budget	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
349	/hash_query_count	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
350	/signature_query_count; ring.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
351	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
352	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
353	lemma counted_rom_programming_loss_term_cardinalityE :	States checked lemma counted_rom_programming_loss_term_cardinalityE. The following proof script must derive this proposition from prior definitions and lemmas.
354	counted_rom_programming_loss_term =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
355	((rom_total_query_budget * rom_total_query_budget) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
356	(rom_signature_query_budget * (rom_hash_query_budget + 1%r)) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
357	(rom_signature_query_budget * (rom_hash_query_budget + 1%r))) /	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
358	challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
359	proof.	Begins the proof script for the preceding proposition.
360	by rewrite counted_rom_programming_loss_termE	Discharges the current goal in one step using the cited simplification, lemma, or automation.

Line	EasyCrypt source	Mathematical reading
361	counted_rom_budget_numeratorE.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
362	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
363	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
364	lemma counted_rom_prequery_reprogramming_term_cardinalityE :	States checked lemma
365	counted_rom_prequery_reprogramming_term =	counted_rom_prequery_reprogramming_term_cardinalityE. The following proof script must derive this proposition from prior definitions and lemmas.
366	((rom_total_query_budget * rom_total_query_budget) +	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
367	(rom_signature_query_budget * (rom_hash_query_budget + 1%r))) /	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
368	challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
369	proof.	Ends the proof; EasyCrypt has accepted all remaining obligations.
370	by rewrite counted_rom_prequery_reprogramming_termE	Discharges the current goal in one step using the cited simplification, lemma, or automation.
371	counted_rom_prequery_reprogramming_budget_numeratorE.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
372	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
373	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
374	lemma counted_rom_programming_loss_term_concreteE :	States checked lemma counted_rom_programming_loss_term_concreteE.
375	counted_rom_programming_loss_term =	The following proof script must derive this proposition from prior definitions and lemmas.
376	37%r / 288230376151711744%r.	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
377	proof.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
378	by rewrite counted_rom_programming_loss_termE	Ends the proof; EasyCrypt has accepted all remaining obligations.
379	/challenge_support_cardinality_lower_bound.	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
380	qed.	States checked lemma
381	<i>blank</i>	counted_rom_prequery_reprogramming_term_concreteE. The following proof script must derive this proposition from prior definitions and lemmas.
382	lemma counted_rom_prequery_reprogramming_term_concreteE :	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
383	counted_rom_prequery_reprogramming_term =	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
384	31%r / 288230376151711744%r.	Ends the proof; EasyCrypt has accepted all remaining obligations.
385	proof.	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
386	by rewrite counted_rom_prequery_reprogramming_termE	States checked lemma
387	/challenge_support_cardinality_lower_bound.	counted_rom_prequery_reprogramming_term_concreteE. The following proof script must derive this proposition from prior definitions and lemmas.
388	qed.	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
389	<i>blank</i>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
390	lemma counted_rom_programming_loss_term_checked_subunit :	Ends the proof; EasyCrypt has accepted all remaining obligations.
391	0%r < counted_rom_programming_loss_term /\	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
392	counted_rom_programming_loss_term < 1%r /\	States checked lemma
393	counted_rom_programming_loss_term =	counted_rom_programming_loss_term_checked_subunit. The following proof script must derive this proposition from prior definitions and lemmas.
394	37%r / 288230376151711744%r.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
395	proof.	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.

Line	EasyCrypt source	Mathematical reading
396	split.	Splits a conjunction or structured goal into independent proof obligations.
397	+ by apply counted_rom_programming_loss_term_positive.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
398	split.	Splits a conjunction or structured goal into independent proof obligations.
399	+ by apply counted_rom_programming_loss_term_subunit.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
400	by apply counted_rom_programming_loss_term_concreteE.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
401	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
402	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
403	lemma counted_rom_support_lower_bound_summary :	States checked lemma counted_rom_support_lower_bound_summary. The following proof script must derive this proposition from prior definitions and lemmas.
404	challenge_support_bits_lower_bound = 58 /\	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
405	challenge_support_cardinality_lower_bound =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
406	288230376151711744%r.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
407	proof.	Begins the proof script for the preceding proposition.
408	split.	Splits a conjunction or structured goal into independent proof obligations.
409	+ by rewrite /challenge_support_bits_lower_bound.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
410	by rewrite /challenge_support_cardinality_lower_bound.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
411	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
412	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
413	lemma counted_rom_support_dominates_budget :	States checked lemma counted_rom_support_dominates_budget. The following proof script must derive this proposition from prior definitions and lemmas.
414	37%r < challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
415	proof. by rewrite /challenge_support_cardinality_lower_bound. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
416	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
417	lemma counted_rom_prequery_reprogramming_support_dominates_budget :	States checked lemma counted_rom_prequery_reprogramming_support_dominates_budget. The following proof script must derive this proposition from prior definitions and lemmas.
418	31%r < challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
419	proof. by rewrite /challenge_support_cardinality_lower_bound. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
420	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
421	lemma counted_rom_prequery_reprogramming_term_subunit :	States checked lemma counted_rom_prequery_reprogramming_term_subunit. The following proof script must derive this proposition from prior definitions and lemmas.
422	counted_rom_prequery_reprogramming_term < 1%r.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
423	proof.	Begins the proof script for the preceding proposition.
424	rewrite counted_rom_prequery_reprogramming_termE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
425	rewrite ltr_pdivr_mulr.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
426	+ by apply challenge_support_cardinality_lower_bound_positive.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
427	by apply counted_rom_prequery_reprogramming_support_dominates_budget.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
428	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
429	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.

Line	EasyCrypt source	Mathematical reading
430	lemma counted_rom_prequery_reprogramming_from_query_budgets_and_support :	States checked lemma counted_rom_prequery_reprogramming_from_query_budgets_and_support. The following proof script must derive this proposition from prior definitions and lemmas.
431	counted_rom_prequery_reprogramming_term =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
432	((rom_total_query_budget * rom_total_query_budget) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
433	(rom_signature_query_budget * (rom_hash_query_budget + 1%r))) /	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
434	challenge_support_cardinality_lower_bound /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
435	((rom_total_query_budget * rom_total_query_budget) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
436	(rom_signature_query_budget * (rom_hash_query_budget + 1%r))) <	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
437	challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
438	proof.	Begins the proof script for the preceding proposition.
439	split.	Splits a conjunction or structured goal into independent proof obligations.
440	+ by apply counted_rom_prequery_reprogramming_term_cardinalityE.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
441	rewrite counted_rom_prequery_reprogramming_budget_numeratorE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
442	by apply counted_rom_prequery_reprogramming_support_dominates_budget.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
443	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
444	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
445	lemma counted_rom_programming_loss_splitE :	States checked lemma counted_rom_programming_loss_splitE. The following proof script must derive this proposition from prior definitions and lemmas.
446	counted_rom_programming_loss_term =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
447	counted_rom_prequery_reprogramming_term +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
448	counted_rom_min_entropy_term.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
449	proof.	Begins the proof script for the preceding proposition.
450	by rewrite /counted_rom_programming_loss_term	Discharges the current goal in one step using the cited simplification, lemma, or automation.
451	/counted_rom_prequery_reprogramming_term; ring.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
452	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
453	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
454	lemma counted_rom_loss_from_query_budgets_and_support :	States checked lemma counted_rom_loss_from_query_budgets_and_support. The following proof script must derive this proposition from prior definitions and lemmas.
455	counted_rom_programming_loss_term =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
456	((rom_total_query_budget * rom_total_query_budget) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
457	(rom_signature_query_budget * (rom_hash_query_budget + 1%r)) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
458	(rom_signature_query_budget * (rom_hash_query_budget + 1%r))) /	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
459	challenge_support_cardinality_lower_bound /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
460	((rom_total_query_budget * rom_total_query_budget) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
461	(rom_signature_query_budget * (rom_hash_query_budget + 1%r)) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
462	(rom_signature_query_budget * (rom_hash_query_budget + 1%r))) <	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
463	challenge_support_cardinality_lower_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
464	proof.	Begins the proof script for the preceding proposition.
465	split.	Splits a conjunction or structured goal into independent proof obligations.
466	+ by apply counted_rom_programming_loss_term_cardinalityE.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
467	rewrite counted_rom_budget_numeratorE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
468	by apply counted_rom_support_dominates_budget.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
469	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
470	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
471	lemma rejection_sampling_loss_term_nonnegative :	States checked lemma rejection_sampling_loss_term_nonnegative. The following proof script must derive this proposition from prior definitions and lemmas.
472	0%r <= rejection_sampling_loss_term.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
473	proof.	Begins the proof script for the preceding proposition.
474	rewrite /rejection_sampling_loss_term.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
475	by apply addr_ge0; [apply fs_with_aborts_reprogramming_term_nonnegative	Discharges the current goal in one step using the cited simplification, lemma, or automation.
476	apply fs_with_aborts_min_entropy_term_nonnegative].	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
477	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
478	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
479	lemma fs_with_aborts_min_entropy_termE :	States checked lemma fs_with_aborts_min_entropy_termE. The following proof script must derive this proposition from prior definitions and lemmas.
480	fs_with_aborts_min_entropy_term = 12%r.	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
481	proof.	Begins the proof script for the preceding proposition.
482	by rewrite /fs_with_aborts_min_entropy_term /min_entropy_loss_factor	Discharges the current goal in one step using the cited simplification, lemma, or automation.
483	signing_query_scaleE /signature_query_count /hash_query_count; ring.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
484	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
485	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
486	lemma fs_with_aborts_min_entropy_term_ge1 :	States checked lemma fs_with_aborts_min_entropy_term_ge1. The following proof script must derive this proposition from prior definitions and lemmas.
487	1%r <= fs_with_aborts_min_entropy_term.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
488	proof.	Begins the proof script for the preceding proposition.
489	by rewrite fs_with_aborts_min_entropy_termE.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
490	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
491	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
492	lemma rejection_sampling_loss_term_ge1 :	States checked lemma rejection_sampling_loss_term_ge1. The following proof script must derive this proposition from prior definitions and lemmas.
493	1%r <= rejection_sampling_loss_term.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
494	proof.	Begins the proof script for the preceding proposition.
495	rewrite /rejection_sampling_loss_term.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
496	smt(fs_with_aborts_min_entropy_term_ge1	Calls SMT automation for first-order arithmetic, list, or algebraic side conditions.
497	fs_with_aborts_reprogramming_term_nonnegative).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
498	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
499	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
500	lemma bimodal_to_msis_reduction_loss_termE :	States checked lemma bimodal_to_msis_reduction_loss_termE. The following proof script must derive this proposition from prior definitions and lemmas.

Line	EasyCrypt source	Mathematical reading
501	<code>bimodal_to_msis_reduction_loss_term = 0%r.</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
502	<code>proof. by rewrite /bimodal_to_msis_reduction_loss_term. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
503	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
504	<code>end HAETAE_Reductions.</code>	Closes the current theory or module block.